

A X I O T I C A I

Intelligence Beyond Scale

A European Research Direction for Frontier AI

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Executive Introduction

The defining question of this moment in artificial intelligence is not whether scaling will continue, but what comes after it. Over the past decade, the dominant paradigm — that capability follows predictably from accumulating parameters, data, and compute — has yielded genuinely remarkable results. Yet the same frontier those systems have opened now exposes their limitations: brittleness under distributional shift, fragility in compositional reasoning, opacity of inference, and an energy and capital intensity that concentrates frontier development among a vanishingly small number of well-capitalised actors. The ceiling of the scaling paradigm may be approaching faster than its proponents anticipated.

Axiotic AI submits this statement from a substantive conviction: that a scientifically grounded alternative to the scaling consensus is not only possible, but strategically timely for the European Union. We hold that intelligence in artificial systems — as in biological ones — emerges from *structure*, *interaction*, and *information*, not from parameter count alone. This thesis is not speculative. It is rooted in decades of converging research across cognitive science, neuroscience, and information theory, and it carries direct implications for how Europe should position itself in the next phase of frontier AI development.

Europe does not lack scientific talent, intellectual tradition, or institutional ambition. What it has lacked is a frontier AI strategy calibrated to its actual strengths. The prevailing paradigm structurally favours actors with deep capital reserves and concentrated infrastructure — neither of which characterises the European research ecosystem. A beyond-scaling direction maps directly onto Europe’s comparative advantages: rigorous interdisciplinary inquiry, open science, constraint-driven innovation, and a regulatory premium on trustworthiness and interpretability. It also speaks to strategic autonomy and digital sovereignty — goals that cannot be achieved while European AI capability depends structurally on architectures and agendas set elsewhere.

Europe’s path to frontier AI leadership runs through architectural innovation, not scale. The window for establishing that position is open. This forum is an appropriate place to begin.

Section 1: A Research Direction Worth Pursuing

1.1 The Biological Challenge to the Scaling Consensus

The adult human brain operates on approximately twenty watts while simultaneously performing perception, language, planning, and abstract reasoning — often in real time, on inputs it has never encountered before. Current large-scale AI systems require orders of magnitude more energy for considerably narrower tasks. This disparity is a scientific puzzle: biological intelligence does not operate as a scaled-up version of what our current architectures do.

Tishby’s *information bottleneck* framework offers one such principle: efficient intelligence is not about storing more information, but compressing representations to retain only what is causally relevant. The brain is an aggressive filter tuned to discard irrelevant detail while preserving structure necessary for generalisation. Lieder and Griffiths’ *resource rationality* extends this further: biological cognition achieves near-optimal decisions under strict computational budgets. Intelligence is defined by how well a system allocates limited resources — not by how many it consumes. The hippocampal-cortical memory system illustrates a third principle: episodic memory is consolidated and abstracted over time, enabling generalisation that static parameter storage does not replicate.

1.2 A Direction Grounded in These Principles

Rather than building larger static parameter banks, the research direction we advance explores architectures that allocate compute dynamically — concentrating inference resources where uncertainty is highest. Rather than flattening time into a fixed context window, it investigates persistent latent memory structures that allow reasoning to deepen. Rather than relying solely on learned statistical co-occurrence,

it pursues relational and compositional representations capable of binding concepts in structured, manipulable ways.

None of these strands has yet produced a system matching the breadth and efficiency of biological intelligence — nor do we claim otherwise. The scientific motivation is sound, the theoretical frameworks are mature, and the engineering challenges, though substantial, are tractable.

1.3 Europe’s Distinctive Opportunity

This direction is strategically aligned with where European research already stands. The cognitive science traditions of Edinburgh, Amsterdam, Paris, and Berlin; the computational neuroscience programmes in London, Zürich, and Stockholm; the information theory lineage running through European universities — these represent genuine and underutilised intellectual assets. Europe’s research culture has long prioritised principled, theoretically grounded inquiry over purely empirical scaling. In a landscape where the limits of scale are becoming apparent, that culture may prove an enduring strength.

The Commission is invited to consider not merely whether Europe can compete *within* the scaling paradigm, but whether an alternative frontier exists — one on which European research is better positioned to lead.

Section 2: European Challenges and the Case for a Different Path

Europe’s position in frontier AI is not one of absence — it is one of structural misalignment. The continent possesses world-class research institutions, deep scientific rigour, and a regulatory framework demanding trustworthiness. Yet these assets have not translated into competitive parity. The reasons are structural, interrelated, and amplified by the prevailing scaling paradigm.

2.1 Compute Access and Capital Intensity

The single greatest bottleneck is compute. Training runs for leading models have required clusters of 10,000–25,000 accelerators operating for months, with capital expenditure of \$50–150 million per run. The US Stargate initiative represents a \$500 billion commitment over four years; individual US hyperscalers invest tens of billions annually. Europe’s compute base — principally EuroHPC — represents meaningful progress but remains structurally insufficient for frontier-scale training. No European AI entity currently operates infrastructure comparable to that routinely deployed by OpenAI, Google DeepMind, or Baidu.

2.2 Energy, Talent, and Fragmented Funding

Energy costs compound the challenge. European industrial electricity costs remain substantially higher than those enjoyed by US and Chinese counterparts, where data centre power is frequently subsidised as an element of national AI strategy.

On talent: Europe produces a globally disproportionate share of AI research graduates, yet a well-documented fraction relocate to US institutions, drawn by salary differentials of three to five times those available in European academia. Europe trains talent that builds US frontier capabilities. Funding fragmentation exacerbates all of the above: investment is split across 27 Member States with partially overlapping, partially contradictory strategies. Horizon Europe provides a valuable transnational layer but is not structured to fund the sustained, risk-tolerant, large-scale development that defines frontier AI.

2.3 How the Scaling Paradigm Amplifies European Disadvantages

The scaling paradigm structurally favours actors with deep capital reserves, large proprietary infrastructure, and monopolistic access to web-scale data. Each successive generation raises the resource threshold for competitive participation — the compute gap between GPT-3 and GPT-4 increased by an estimated

factor of one hundred or more. Winner-takes-most dynamics emerge; capital compounds, talent concentrates, and research is shaped by who can afford it. There is also a security dimension: a Union whose critical AI infrastructure is trained on architectures designed abroad, run on hardware manufactured abroad, and hosted on clouds owned abroad faces structural dependency that carries acute strategic risk in scenarios of geopolitical tension.

2.4 How a Beyond-Scaling Direction Partially Mitigates Them

Architectures that allocate compute selectively can achieve capable behaviour within substantially smaller resource envelopes, bringing meaningful research within the range of European infrastructure. The recent demonstration by DeepSeek — that highly competitive frontier performance can be achieved at a fraction of US training costs — provides empirical evidence that efficiency-oriented research is a legitimate — not consolatory — path to frontier capability. Europe’s structural constraints, reframed, become a driver of scientific innovation rather than merely a liability.

Section 3: Strategic Recommendations

Part I: Actions for the European Commission

Recommendation 1 — Establish a dedicated funding strand for architectural diversity in AI

The Commission should create a ring-fenced funding stream within Horizon Europe targeting AI architectures that depart from transformer-based scaling — including dynamic attention mechanisms, persistent latent memory systems, relational and compositional learning, world-grounded multimodal architectures, and neuroscientifically-inspired temporal reasoning. A minimum allocation of €500 million over 2026–2030, structured as four-to-six-year grants, is proposed. Short grant cycles have historically disadvantaged foundational AI research in favour of incremental applied work; this instrument must be designed to resist that bias.

Recommendation 2 — Mandate efficiency-aware evaluation in publicly funded AI procurement and benchmarking

The Commission, through its AI Office and the AI TEFs, should mandate that all publicly funded AI systems be evaluated not solely on capability benchmarks but on capability-per-unit-compute and capability-per-unit-energy metrics. ENISA and the Joint Research Centre should publish, by Q4 2026, a reference framework for efficiency-aware AI evaluation. By making efficiency a first-class criterion, the Commission creates market conditions that reward the architectural approaches described in Section 1 and removes the institutional bias toward scale-maximising systems that currently shapes both procurement and grant assessment.

Recommendation 3 — Build open, shared model infrastructure under a European governance structure

The Commission should fund a European Open AI Infrastructure: open-weight foundational model releases, training checkpoints, and intermediate-representation artefacts, hosted under a neutral European governance body and accessible to academic and small commercial entities without charge. European researchers on novel architectures cannot currently validate approaches at meaningful scale. A shared European base removes this barrier. Beyond the research rationale, it constitutes a concrete exercise of digital sovereignty: ensuring foundational AI artefacts exist and persist under European governance, regardless of commercial decisions by non-European platform providers.

Part II: Actions for Member States

Recommendation 4 — Reserve compute capacity for non-scaling-paradigm research

Member States operating or co-funding AI compute facilities — including those under EuroHPC — should introduce dedicated allocation pools reserved for beyond-scaling research, sized at a minimum of 15–20% of total national AI compute capacity, governed by panels that include architectural diversity as

a formal criterion. Novel-paradigm teams cannot currently compete for general allocation against well-resourced transformer-scaling projects. A reserved allocation corrects this structural disadvantage.

Recommendation 5 — Anchor talent retention through national research chairs and secondment programmes

Member States should establish co-funded national AI Research Chair programmes targeting mid-career researchers in architectural AI, cognitive-inspired computing, and information-theoretic learning. Chair holders should carry protected research time of at least 60%, industry-benchmarked salaries, and access to the compute reservations in Recommendation 4. Complementary secondment schemes enabling movement between European academic institutions and European companies will reduce structural incentives to relocate to North American hyperscaler research divisions.

Part III: Actions for European Actors in Industry and Academia

Recommendation 6 — Adopt paradigm diversity as a founding principle of European AI consortia

European AI research consortia submitting to EU funding calls should formally commit to paradigm diversity as a scientific principle — analogous to commitments to open science or gender balance. In practice, proposals for large-scale AI research projects should include at least one architectural workstream that does not presuppose transformer-based scaling as the terminal design space. A research ecosystem converging entirely on one architectural class is fragile — scientifically, economically, and strategically.

Recommendation 7 — Establish a European AI Architecture Observatory

European academic institutions, in partnership with the Commission's AI Office, should co-establish an observatory charged with continuously surveying the global landscape of AI architectural research, tracking efficiency benchmarks across paradigms, and publishing annual comparative assessments accessible to policymakers and funders. This observatory provides the evidentiary basis for the evaluation framework in Recommendation 2 and furnishes Member States with the intelligence required to calibrate compute reservation policies under Recommendation 4.

Conclusion

Europe's competitiveness in AI will not be determined in the present moment — it will be determined in the transition between paradigms. The current paradigm was established under conditions that structurally favour large US and Chinese actors. Europe did not design those conditions and cannot win on those terms alone.

The next paradigm is being written now. If intelligence in machines is, as the evidence increasingly suggests, more a function of architecture and structure than of scale, then the competitive landscape resets — and in that reset, Europe's assets become advantages rather than handicaps. The recommendations above are not defensive measures. They are the scaffolding for an offensive strategy: to fund the research direction, build the shared infrastructure, retain the talent, and establish the evaluation standards that position Europe not as an importer of the next AI paradigm, but as its originator.

The scientific case is substantive. The strategic alignment is clear. The window is open. We urge the Commission to act accordingly.

Axiotic AI is a beyond-scaling AI research laboratory and consultancy. Our research thesis — that intelligence emerges from structure, interaction, and information — informs both our scientific programme and our engagement with European frontier AI policy. We welcome dialogue with the Commission and its advisors on any aspect of this submission.